

BALASAKTHI SHANMUGARAJA

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[Linkdin](#)

PROFILE

Telecommunications Engineering student at Politecnico di Milano with academic and project experience in mobile radio networks, RF systems, wireless communications, network planning and IoT. Hands-on experience with Python-based network modelling and optimization. Strong interest in 4G/5G mobile networks, IMS, VoLTE/VoWiFi, mobile device testing and network troubleshooting. Seeking an internship where I can apply my telecommunications background to mobile network testing, technical support and wireless technology development.

TECHNICAL SKILLS

Programming & Tools: Python, C++, MATLAB, NumPy, SQL, Matplotlib, Linux, Wireshark, Git, Microwave Office

Mobile Networks & IoT: Wireless Network Planning, Radio Propagation, TCP/IP, Wireless Sensor Networks, ESP32, ESP-NOW, IoT

Network Analysis & Emerging Technologies: MILP, Network Planning Optimization, Technical Simulation, Data Visualization, IMS, VoLTE, VoWiFi, VoNR, JIRA

EDUCATION

MSc in Telecommunications Engineering

2025 – Present

Politecnico di Milano

Track communication network and internet

Average: 24.65/30

Coursework:

- Mobile Radio Networks
- Wireless Networks
- RF Systems
- Network Measurement and Data Analysis
- Multimedia Internet
- Computing Infrastructures
- Internet of Things
- Network Cryptography
- Localization, Navigation and Smart Mobility

B.E. in Electronics and Communication Engineering
NPR College of Engineering and Technology
CGPA: 7.52/10
Thesis: hybrid vehicle with motion sensors

2020 – 2024

ACADEMIC PROJECTS

Mobile Radio Network Planning & Optimization

Python, AMPL, HiGHS, NumPy, Matplotlib

Developed a MILP-based network planning model to optimize macro and micro base-station deployment considering coverage, throughput and deployment cost. Modelled radio propagation using Okumura-Hata and evaluated network performance using Python.

RF Front-End Design & Performance Analysis

AWR Microwave Office/VSS

Designed and simulated an RF receiver front-end including LNA, mixer, filters and local oscillator at 4.25 GHz. Evaluated S-parameters, impedance matching, gain, noise figure, IP3, P1dB and EVM, including phase-noise and interference effects.

Energy-Efficient IoT Wireless Sensor Node

ESP32, ESP-NOW, HC-SR04

Developed an ESP32-based wireless parking occupancy sensor using ultrasonic sensing and ESP-NOW communication. Implemented deep sleep and duty-cycle optimization and evaluated energy consumption and estimated battery lifetime.

Microwave Network Hardware Failure Classification

Python, Pandas, Scikit-learn, XGBoost, SMOTEN, CVAE

Developed a machine-learning pipeline to classify microwave network hardware failures using real and synthetic data. Applied Random Forest, XGBoost, SMOTEN and CVAE and evaluated performance using precision, recall and F1-score.

LLM-Based Automated Question Answering System

Python, PyTorch, Hugging Face, Qwen2.5-7B

Developed an automated question-answering pipeline using Qwen2.5-7B-Instruct, integrating API-based question retrieval and automated answer processing. Implemented 4-bit quantization for efficient GPU inference and evaluated model performance across different question categories.